Statistical foundations of business analytics

Day 4 Time series and forecasting baselines

Daily orders, simple forecasts and honest evaluation

29 October 2026
Lecture slides for a 150 minute session

The 150 minute lecture route

Elapsed time	Topic
0 to 10	Business decision and forecast task
10 to 30	Metric, calendar and visible patterns
30 to 45	Lags, stationarity and differencing
45 to 65	Four simple forecasting baselines
65 to 70	Five minute break
70 to 90	Earlier validation and forecast accuracy
90 to 110	Uncertainty and capacity decisions
110 to 140	Worked forecasting exercises
140 to 150	Review and four-day synthesis

Optional formulas and spreadsheet examples follow the core lecture.

A weekly total and a daily staffing decision

ShopEase expects about 1,200 paid orders next week. The packing team needs a capacity plan for each day.

Is the weekly total enough to schedule daily shifts? What additional forecast would help?

Daily shifts need the daily pattern

The same weekly total could be spread evenly or concentrated on Saturday.

Daily staffing needs seven daily forecasts, together with uncertainty and the cost of insufficient capacity.

Overpredicting Monday and underpredicting Saturday can cancel in a weekly total while causing problems on both days.

The decision determines the forecast target

Decision	What is forecast	How far ahead
Daily packing shifts	New paid orders each day	Seven daily values
Packaging purchase	Total orders over a week	One seven-day total
Cash planning	Monthly net cash receipts	Next three months
Support staffing	Incoming calls each hour	Next day by hour

The target, time interval, issue time and horizon belong to the forecast definition.

The four business questions

Day	Question
1 Experiments	Does an intervention change a defined metric?
2 Association rules	Which products appear together?
3 Time to event	How long until an individual first experiences an event?
4 Forecasting	What will the calendar-indexed metric be in future periods?

A sales forecast under expected conditions does not establish the causal effect of a promotion.

The main series and its forecast contract

Field	Definition
What is counted	Eligible new paid orders, one per distinct order ID
Time interval	Calendar day, including weekends
Day boundaries	Asia/Kolkata time zone
When issued	End of day 28, a Sunday, after its count is available
Future targets	Days 29 to 35, Monday through Sunday
Business action	Daily processing capacity for the coming week

A customer with three eligible orders contributes three orders.

What is known when Sunday ends

The team knows next week's calendar and has a planned promotion. Next week's realized website traffic will arrive later.

Which inputs can a forecast issued this Sunday actually use? What about later refunds that revise old counts?

Only information available at the issue time

Future weekdays are known. A planned promotion is usable if its plan was available then, with implementation uncertainty stated.

Realized future traffic is unavailable. Historical counts revised later may also differ from what the team knew on Sunday.

Historical evaluation should reproduce the original information cutoff. Keep historical snapshots when revisions matter.

Zero orders and a missing daily extract

A reliable system recorded zero orders on Tuesday. Wednesday's extract failed. Thursday is still in progress.

Should all three dates enter the completed-day series as zero?

Zero is known and missing is unknown

Situation	Initial treatment
Tuesday: reliable zero count	Retain the observed zero
Wednesday: failed extract	Mark missing and investigate
Thursday: incomplete day	Exclude from completed-day training or model explicitly

Construct the expected daily calendar before replacing gaps. Record closures and deduplicate repeated order events.

Pooling two daily conversion rates

Day A has 10 paid orders from 100 sessions. Day B has 90 paid orders from 300 sessions.

The daily rates are 10% and 30%. Is the combined conversion rate their average, 20%?

The pooled conversion rate is 25%

$$\text{Combined conversion} = \frac{10 + 90}{100 + 300} = \frac{100}{400} = 25\%$$

The two days contribute different numbers of sessions. Pool the order counts and session counts before dividing.

A rate forecast can hide changes in traffic volume. Orders, revenue and customers are different targets.

The quantity determines the calendar calculation

Daily order arrivals are a flow. Adding seven days gives the weekly order total.

End-of-day inventory is a stock. Adding seven snapshots does not give inventory at the end of the week.

Lag 7 means seven observations earlier. It means one calendar week only when every calendar day appears in the daily series.

An unusually low sales day

A product sells very little on a day when it is out of stock. Another day has a large unexplained fall.

Do recorded sales necessarily measure underlying demand?
Should the analyst delete unusual days?

Observed sales can be limited by availability

Stockouts can prevent customers from placing orders. Recorded sales then measure a constrained outcome.

Investigate outages, stockouts, tracking failures and business events. Record what is known about the date.

Deleting a day only because it worsens accuracy can remove a real operational risk.

The main example has 28 observed daily counts

Weekday	Week 1	Week 2	Week 3	Week 4
Monday	100	110	120	130
Tuesday	120	130	140	150
Wednesday	130	140	150	160
Thursday	140	150	160	170
Friday	160	170	180	190
Saturday	200	210	220	230
Sunday	150	160	170	180

Four completed weeks, Monday to Sunday. The last observed day is 28.

What pattern appears in the four observed weeks

Saturday counts are 200, 210, 220 and 230. Monday counts are 100, 110, 120 and 130.

What repeats within a week? What changes from one week to the next?

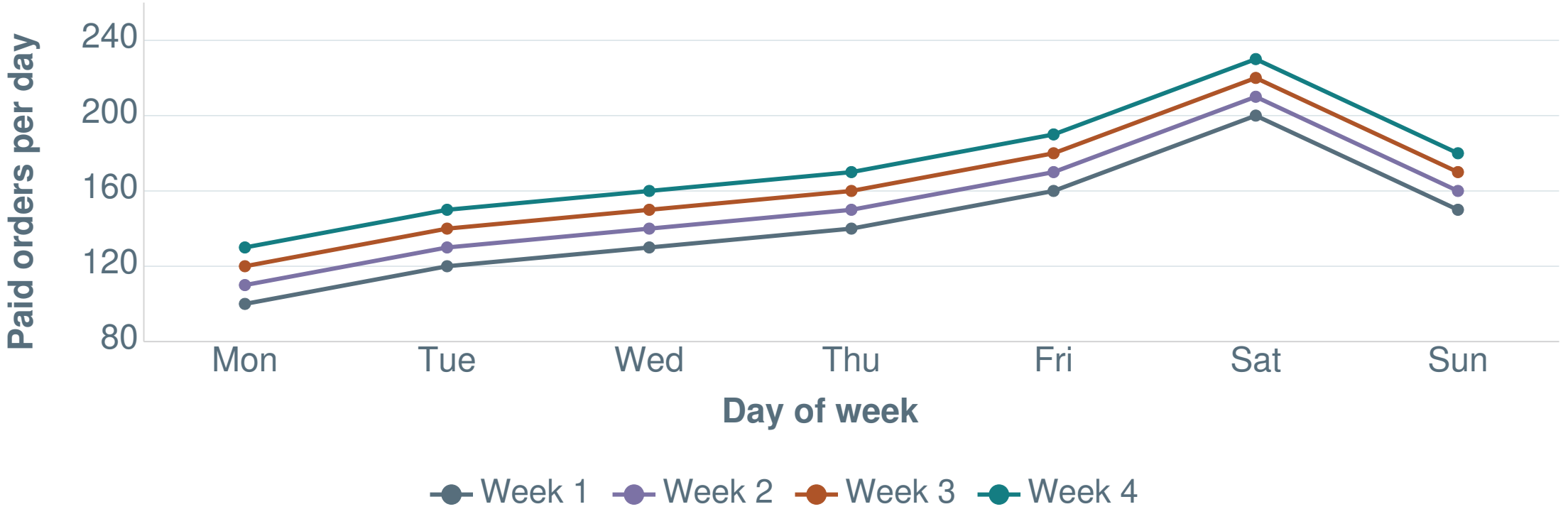
The weekday shape repeats while the level rises

Saturdays are high and Mondays are low. That recurring calendar shape suggests weekly seasonality.

Each weekday increases by ten orders from one week to the next. The whole pattern shifts upward.

This is a deliberately regular teaching dataset. It does not prove that future counts will keep increasing by ten.

The same weekday pattern shifts upward



Compare like weekdays across weeks. Monday-to-Saturday movement within a week also reflects the weekday pattern.

The totals behind the training averages

Quantity	Calculation	Result
Week 1 total	100 + 120 + 130 + 140 + 160 + 200 + 150	1,000
Week 2, 3 and 4 totals	1,070, 1,140 and 1,210	
All 28 days	1,000 + 1,070 + 1,140 + 1,210	4,420
Full-history daily mean	4,420 / 28	157.857
Latest-week daily mean	1,210 / 7	172.857

Both averages count orders per day, but they summarize different historical windows.

Patterns an analyst may try to represent

Pattern	Plain-language meaning	Example
Changing level or trend	Persistent movement over time	Growing customer base
Seasonality	Repetition at a known calendar frequency	Weekday pattern
Cycle	Fluctuation without a fixed seasonal period	Business expansion and contraction
Irregular remainder	Variation left after represented structure	Unexplained daily variation

Absolute changes and proportional changes

An additive description combines an evolving level, a seasonal contribution and remaining variation.

$$y_t = L_t + S_t + R_t$$

A multiplicative description makes the variation proportional to the level.

$$y_t = L_t S_t R_t$$

The main data have a stable absolute weekday shape. We do not fit a formal decomposition here.

A festival spike and a seven-day seasonal pattern

A large festival changes date from year to year. ShopEase also has a recurring Saturday peak.

Will copying the same weekday last week automatically reproduce the festival effect?

A calendar event may need its own treatment

A seven-day rule represents a recurring weekday pattern. A moving holiday may need explicit calendar information or scenarios.

Trend and seasonality can coexist. Copying last week may reproduce its shape while missing growth.

Daily series can also contain annual patterns. Four weeks cannot establish those patterns.

A lag is an earlier observation

In a complete daily calendar, lag 1 means yesterday and lag 7 means the same weekday last week.

A lag plot pairs each current value with the value a chosen number of observations earlier.

The time index carries information. Shuffling rows changes those relationships.

Finding the value seven days earlier

Week 2 Monday is day 8 with 110 orders. Week 3 Saturday is day 20 with 220 orders.

Which observation is lag 7 for each? What is the difference from the earlier value?

Lag 7 compares the same weekday

Day 8 Monday pairs with day 1 Monday: 110 - 100 = 10 orders.

Day 20 Saturday pairs with day 13 Saturday: 220 - 210 = 10 orders.

Every available lag-7 pair in these training data differs by ten.

The current count is ten above its lag-7 value



Each point pairs a day with the same weekday one week earlier. Some pairs overlap exactly.

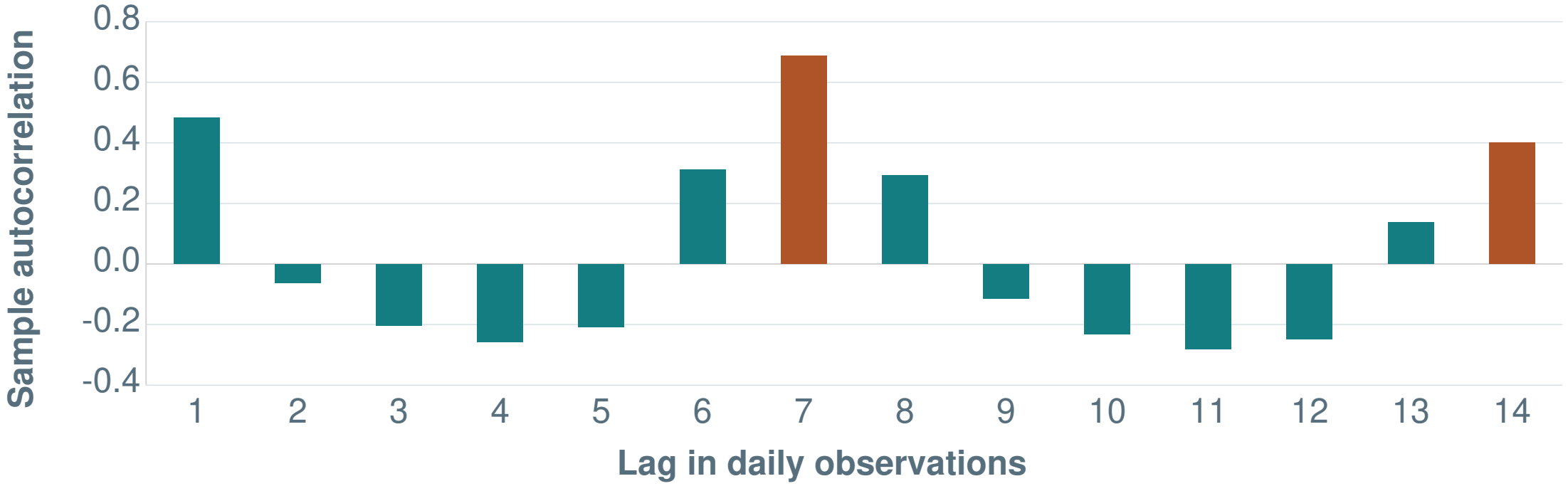
Autocorrelation summarizes relationships at a lag

Autocorrelation compares a series with its own earlier values. The ACF displays this summary over several lags.

A large lag-7 coefficient may suggest useful weekly structure. Trend can also create strong raw-series relationships.

Association over time does not show that yesterday's orders cause today's orders.

The raw series has visible lag-7 and lag-14 structure



Lag 7 is 0.688 and lag 14 is 0.400 under the notes' sample ACF formula. Four cycles provide limited evidence.

Stationarity describes stable statistical behaviour

Weak stationarity means a constant mean and constant finite variance. Covariance depends on the lag rather than the calendar date.

The observed values may still fluctuate. Stationarity does not require every value to be equal.

Our raw example changes level and has a systematic weekday pattern, so a constant-mean stationary description is unsuitable.

Adjacent-day changes in the first week

The first week has Monday 100, Tuesday 120, Wednesday 130, Saturday 200 and Sunday 150 orders.

Calculate Tuesday minus Monday, Wednesday minus Tuesday and Sunday minus Saturday.

First differences describe change rather than level

$$\Delta y_t = y_t - y_{t-1}$$

Tuesday: 120 - 100 = 20.
Wednesday: 130 - 120 = 10.
Sunday: 150 - 200 = -50.

Adjacent-day changes still reflect the weekday pattern. A negative Sunday change does not by itself imply a sustained decline.

Changes from the same weekday last week

Week 2 Monday is 110 and week 1 Monday is 100. Week 2 Saturday is 210 and week 1 Saturday is 200.

What does subtracting the lag-7 value show?

Every available seasonal difference equals ten

$$\Delta_7 y_t = y_t - y_{t-7}$$

Monday: 110 - 100 = 10. Saturday: 210 - 200 = 10. The other corresponding weekdays behave the same way.

Subtracting matching weekdays removes the repeated weekday shape and exposes the weekly increment built into these data.

A perfectly regular example has limits

The seasonal differences here are constant by construction. Their sample variance is zero, so their ACF is undefined.

A transformation can clarify a pattern without proving that the pattern will continue.

Differencing is method-dependent. Seasonal naive forecasts can use the original counts directly.

Unit-root tests and formal ARIMA identification lie beyond this session.

When issued and how far ahead

Target day	Day of week	Days ahead	Actual data available
29	Monday	1	Days 1 to 28
30	Tuesday	2	Days 1 to 28
35	Sunday	7	Days 1 to 28

Forecast origin = issue time. Horizon = periods ahead.

All seven forecasts in this task are issued together at the end of Sunday, day 28.

Can Tuesday use Monday's actual count

At the end of day 28, the latest count is 180. Monday later records 140 orders.

Should we replace Sunday's Tuesday naive forecast with 140 when evaluating the original seven-day forecast?

An updated forecast has a different issue time

The original Sunday-issued naive forecast is 180 for both Monday and Tuesday.

After Monday ends, a new one-step forecast for Tuesday can use Monday's 140 orders.

Replacing the original prediction changes the task. Compare methods under the same issue schedule and horizon.

The notation records the information cutoff

$$\widehat{y}_{T+h|T}$$

T is the last observed period. T + h is the future target period. The part after the vertical bar records the information cutoff.

$$T = 28, \qquad h = 1, \ldots, 7, \qquad m = 7$$

m is the seasonal period. Monthly data with annual repetition have m = 12, even when only three months are forecast.

Four simple forecasting baselines

Mean: repeat the average of all training observations.

Naive: repeat the latest observed value.

Seasonal naive: repeat the latest observed seasonal cycle.

Drift: extend the average change between the first and last observations.

These are reference methods against which additional complexity must earn its value.

Mean forecast: repeat the historical average

$$\overline{y} = \frac{4{,}420}{28} = 157.857\ldots$$

Every future day receives 157.857 orders, including Monday and Saturday.

The method uses the average historical level. It ignores the weekday pattern and can lag behind growth.

Keep the unrounded value for scoring. Convert forecasts to integer capacity later under a decision rule.

Naive forecast: repeat the latest observation

$$\widehat{y}_{28+h|28} = y_{28} = 180$$

All seven future days receive 180, the last observed Sunday count.

The method carries the latest value forward. It still misses the recurring weekday pattern.

An unusual final day or data error can make that reference poor. A flat point forecast can still have substantial uncertainty.

A separate four-observation example

$$8, \quad 10, \quad 9, \quad 13$$

Using only these observations, what do the mean and naive methods forecast for the next two periods?

The mean gives ten and naive gives thirteen

$$\text{Mean} = \frac{8 + 10 + 9 + 13}{4} = 10$$

Mean forecasts: 10 and 10. Naive forecasts: 13 and 13.

Both rules use the same history but summarize it differently.

Seasonal naive: copy the matching day last week

Next Monday copies the last observed Monday. Next Saturday copies the last observed Saturday.

$$\widehat{y}_{T+h|T} = y_{T+h-m}, \qquad 1 \leq h \leq m$$

For Monday: 28 + 1 - 7 = day 22, with 130 orders. For Saturday: 28 + 6 - 7 = day 27, with 230 orders.

The formula shown covers one seasonal cycle.

The exact observation used for each forecast

Future day	Weekday	Earlier source day	Forecast
29	Monday	22	130
30	Tuesday	23	150
31	Wednesday	24	160
32	Thursday	25	170
33	Friday	26	190
34	Saturday	27	230
35	Sunday	28	180

The forecast weekly total is 1,210. This repeats week 4 and adds no growth adjustment.

The Monday two weeks ahead

The last observed Monday is day 22 with 130 orders. All forecasts must use only days 1 to 28.

What does seasonal naive forecast for day 36, the Monday two weeks ahead?

The last observed week repeats again

The forecast is 130 again. The most recent observed Monday remains day 22.

Day 29 is still a future actual at the original issue time. It cannot supply an observed input for day 36.

Updating after another week creates a new forecast origin.

Drift: average change between the endpoints

The first observation is 100 and the last is 180. There are 27 one-day intervals between 28 observations.

$$\text{Drift per day} = \frac{180 - 100}{28 - 1} = \frac{80}{27} = 2.963\ldots$$

This is an endpoint slope. It differs from fitting a least-squares line to all observations.

The endpoints fall on different weekdays, so the slope mixes trend with seasonal position.

Drift extends the latest count by that daily change

$$\widehat{y}_{28+h|28} = 180 + h\frac{80}{27}$$

Horizon	Calculation	Forecast
1 day: Monday	180 + 1 x 80 / 27	182.963
2 days: Tuesday	180 + 2 x 80 / 27	185.926
7 days: Sunday	180 + 7 x 80 / 27	200.741

The trajectory increases smoothly and misses the weekday shape.

Drift in the four-observation example

$$8, \quad 10, \quad 9, \quad 13$$

How many intervals separate the endpoints? Calculate the next one-step and two-step drift forecasts.

Five units of change over three intervals

$$\text{Daily drift} = \frac{13 - 8}{4 - 1} = \frac{5}{3}$$

$$\widehat{y}_{5|4} = 13 + \frac{5}{3} = 14.667\ldots$$

$$\widehat{y}_{6|4} = 13 + 2 \times \frac{5}{3} = 16.333\ldots$$

The denominator counts intervals, not observations.

Seven forecasts issued together after day 28

Weekday	Mean	Naive	Seasonal naive	Drift
Monday	157.857	180	130	182.963
Tuesday	157.857	180	150	185.926
Wednesday	157.857	180	160	188.889
Thursday	157.857	180	170	191.852
Friday	157.857	180	190	194.815
Saturday	157.857	180	230	197.778
Sunday	157.857	180	180	200.741

Values are rounded for display. Every method uses only the 28 training observations.

Five minute break

Mean repeats the training average. Naive repeats the latest observation.

Seasonal naive repeats the latest seasonal cycle. Drift extends the endpoint average change.

The next question is how to compare them using honest forecasts of later observations.

Rehearsing the real forecasting task

Treat an earlier Sunday as if it were the current issue time. Train on the observations available then and predict the next seven days.

Compare those predictions with the later actual counts. Repeat at another Sunday before selecting the method.

Randomly mixing dates can let future conditions inform predictions of earlier dates.

Earlier practice windows and the final test week

Issue time	Available training days	Future days scored	Purpose
End of day 14	1 to 14	15 to 21	Compare methods
End of day 21	1 to 21	22 to 28	Compare methods
End of day 28	1 to 28	29 to 35	Final test of selected rule

Training expands as observations become available. Method selection ends before the final test outcomes are inspected.

One earlier Sunday in detail

At the end of day 14	What the method uses	Forecast for week 3
Mean	Weeks 1 and 2 total 2,070	147.857 each day
Naive	Day 14 Sunday has 160 orders	160 each day
Seasonal naive	Week 2 counts	110, 130, 140, 150, 170, 210, 160
Drift	160 + h x (160 - 100) / 13	164.615 to 192.308

Week 3 outcomes are unavailable to these forecasts. They enter only when scoring.

Forecast error means actual minus forecast

$$e_i = y_i - \widehat{y_i}$$

Actual	Forecast	Error	Meaning
140	130	+10	Forecast was ten too low
140	180	-40	Forecast was forty too high

With this convention, positive error means underprediction.

Average absolute error in the first practice week

At origin 14, naive predicts 160 for every day of week 3. Week 3 actuals are 120, 140, 150, 160, 180, 220 and 170.

Quantity	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Actual minus 160	-40	-20	-10	0	20	60	10
Absolute error	40	20	10	0	20	60	10

$$\text{MAE} = \frac{40 + 20 + 10 + 0 + 20 + 60 + 10}{7} = 22.857\ldots$$

Seasonal naive is ten low on every day of this practice week, so its MAE is ten.

Seasonal naive has the smallest earlier validation MAE

Method	Days 15 to 21	Days 22 to 28	Average MAE
Mean	25.204	27.347	26.276
Naive	22.857	22.857	22.857
Seasonal naive	10.000	10.000	10.000
Drift	24.835	23.000	23.918

MAE is in orders per day. Equal-length windows allow their MAEs to be averaged directly.

Select seasonal naive here, then refit it on days 1 to 28. Two origins are a teaching illustration, not robust operational evidence.

Can a validation week become later training data

Days 15 to 21 are outcomes scored after the day-14 forecast. The next forecast is issued at the end of day 21.

Can that next forecast train on days 15 to 21? Can it train on day 29?

Availability changes as the issue time advances

Days 15 to 21 have become observed history by the end of day 21. The next training window can include them.

Day 29 remains in the future. Including its actual count would leak future information.

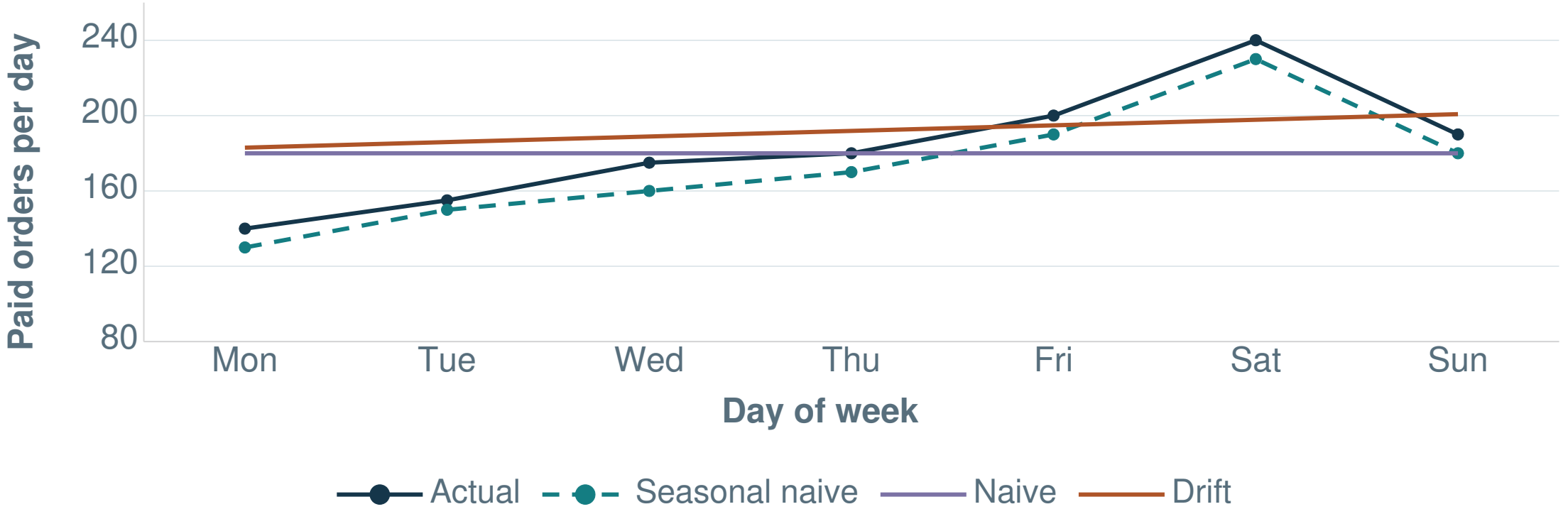
An expanding window keeps older data. A fixed-length rolling window retains a recent history of declared length.

The later week used for final testing

Weekday	Actual	Naive	Seasonal naive	Drift
Monday	140	180	130	182.963
Tuesday	155	180	150	185.926
Wednesday	175	180	160	188.889
Thursday	180	180	170	191.852
Friday	200	180	190	194.815
Saturday	240	180	230	197.778
Sunday	190	180	180	200.741

All forecasts use days 1 to 28. Later actuals score the frozen forecasts.

The selected forecast follows the weekday shape



All seasonal naive forecasts are below the actuals. Its rule carries last week's pattern forward without adding growth.

The seven seasonal naive errors

Weekday	Actual	Forecast	Actual minus forecast
Monday	140	130	10
Tuesday	155	150	5
Wednesday	175	160	15
Thursday	180	170	10
Friday	200	190	10
Saturday	240	230	10
Sunday	190	180	10

Weekly total: forecast 1,210, actual 1,280, shortfall 70 orders.

How large is the average absolute miss

$$e = (10,\ 5,\ 15,\ 10,\ 10,\ 10,\ 10)$$

Take each error's absolute value, add the seven values and divide by seven. What is the MAE, and what are its units?

Mean absolute error: ten orders per day

$$\text{MAE} = \frac{|10| + |5| + |15| + |10| + |10| + |10| + |10|}{7}$$

$$\text{MAE} = \frac{70}{7} = 10 \text{ orders}$$

The average absolute daily miss is ten orders over these seven forecasts.

A measure that gives large errors more weight

$$e = (10,\ 5,\ 15,\ 10,\ 10,\ 10,\ 10)$$

Square each error, average the squares, then take the square root. Why might this differ from MAE?

Root mean squared error: 10.351 orders

$$\sum e_i^2 = 100 + 25 + 225 + 100 + 100 + 100 + 100 = 750$$

$$\text{RMSE} = \sqrt{\frac{750}{7}} = 10.351\ldots \text{ orders}$$

Squaring gives the 15-order error more weight than the 5-order error. The final square root restores the order units.

Can a small signed average hide large misses

Naive forecasts 180 every day. Its signed errors are -40, -25, -5, 0, 20, 60 and 10.

Their sum is only 20. Does dividing by seven establish that the forecast is accurate?

Overprediction and underprediction can cancel

$$\text{Mean error} = \frac{20}{7} = 2.857\ldots$$

$$\text{MAE} = \frac{40 + 25 + 5 + 0 + 20 + 60 + 10}{7} = 22.857\ldots$$

The small signed average hides much larger daily misses. Seasonal naive has mean error ten but lower MAE.

The four baselines on the final test week

Method	MAE	RMSE	Mean error
Mean	30.918	39.038	25.000
Naive	22.857	30.119	2.857
Seasonal naive	10.000	10.351	10.000
Drift	22.540	26.887	-8.995

All entries use orders as the target unit. Positive mean error indicates underprediction.

These later results describe the test. They do not reopen the earlier method-selection decision.

Percentage summaries need a defined denominator

Mean absolute percentage error (MAPE) averages each period's absolute percentage miss. A zero actual makes the usual formula undefined.

$$\text{WAPE} = 100\frac{\sum |e_i|}{\sum y_i} = 100\frac{70}{1{,}280} = 5.469\%$$

Weighted absolute percentage error (WAPE) here divides total absolute error by total actual orders. It is not each day's percentage error.

Its denominator must be positive and changes across evaluation periods.

A zero actual in a percentage error calculation

A separate two-day example has actual counts 0 and 5, with forecasts 2 and 4.

Calculate MAE. Can ordinary MAPE be calculated for both days?

MAE is 1.5 and ordinary MAPE is undefined

$$e = (0 - 2,\ 5 - 4) = (-2,\ 1)$$

$$\text{MAE} = \frac{2 + 1}{2} = 1.5$$

The first MAPE term would divide by the actual value zero. Silently omitting it changes the evaluation.

The pattern of errors guides the next investigation

Observed error pattern	Useful follow-up
Mostly positive errors	Check whether the forecast level is too low
Repeated misses on one weekday	Check the weekday representation
Larger misses at longer horizons	Evaluate each horizon separately
A sustained sudden shift	Investigate process changes and measurement
Larger errors at high volume	Check variance and interval assumptions

A pattern is a diagnosis to investigate, not a demonstrated causal explanation.

Historical fitted residuals and genuine forecast errors

A fitted residual compares an observation with a model fitted on a historical training sample. Parameter estimation can use later observations within that sample.

A genuine forecast error compares an actual with a prediction issued before that actual arrived.

Low training error does not establish future forecasting performance. Uncorrelated residuals alone do not prove a correct model.

Adding ten after inspecting the test errors

All seven seasonal naive test errors are positive. The analyst now adds ten to each original forecast and reports the improved score on the same week.

Is this still an untouched test of the revised rule? What should happen next?

The revised rule needs a later evaluation

The analyst used the test outcomes to design the correction. The adjusted score is a result on inspected development data.

Develop a level-adapting candidate using earlier validation, or treat this week as development data and test the revision on later untouched observations.

The earlier windows already suggested underprediction. Preserve the original issued forecasts and test report.

A forecast value and uncertainty about the outcome

A point forecast summarizes possible future outcomes. A prediction interval describes uncertainty about a specified future observation.

A confidence interval describes uncertainty in a mean or parameter. A prediction interval concerns a future observed value.

$$\text{Approximate } 95\% \text{ interval} = \widehat{y}_{T+h|T} \pm 1.96\widehat{\sigma}_h$$

The normal approximation uses forecast-error standard deviation at horizon h, not the standard error of the historical mean.

A separate random-walk illustration

The latest value is 100. Each future change is independent and normal, with mean zero and standard deviation ten orders.

The point forecast stays 100. What are the one-day and four-day approximate 95% prediction intervals?

Standard deviation at horizon h = 10 times the square root of h.

Uncertainty accumulates across future changes

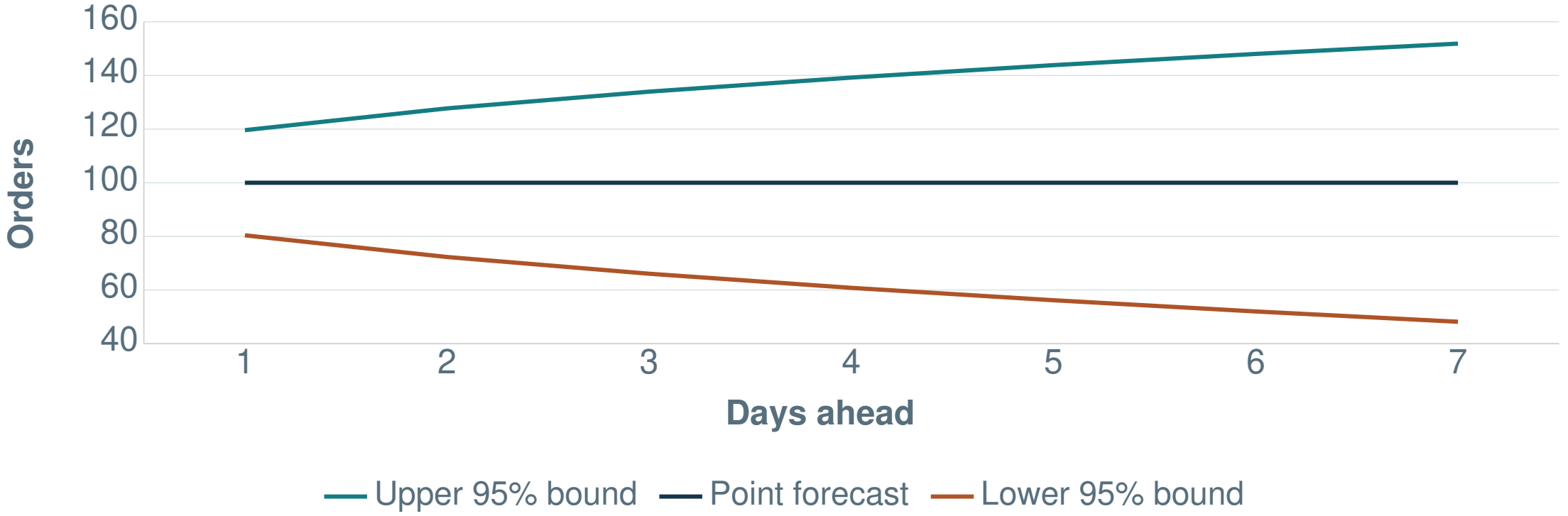
$$\sigma_h = 10\sqrt{h}$$

$$h = 1: \quad 100 \pm 1.96(10) = [80.4,\ 119.6]$$

$$h = 4: \quad 100 \pm 1.96(20) = [60.8,\ 139.2]$$

The point forecast remains flat while the interval widens. These assumptions do not describe an interval fitted to ShopEase.

A flat point forecast can have widening uncertainty



Separate random-walk illustration: standard deviation ten per independent daily increment. Bounds are approximate marginal 95% limits.

Seven separate daily 95% intervals

An analyst supplies a 95% prediction interval for each day of the coming week.

Does that automatically give a 95% chance that all seven daily observations fall inside their intervals?

Daily coverage and whole-week coverage differ

Each daily interval concerns one future target. Simultaneous coverage of all seven targets is a different requirement.

Check coverage and interval width over many relevant later forecasts. A small example cannot establish calibration.

The square-root-of-h rule is model-specific. Low counts, skewness and changing variance may require another predictive distribution.

Capacity when shortages cost more than surplus

Each unit of unused capacity costs two hypothetical cost units. Each order above capacity costs ten.

Under the stated linear shortage and surplus costs, which predictive percentile should capacity target?

Does an 83.3rd percentile mean adding 83.3% to the point forecast?

The cost ratio selects the 83.3rd percentile

$$q = \frac{C_{\text{under}}}{C_{\text{under}} + C_{\text{over}}} = \frac{10}{10 + 2} = 0.8333\ldots$$

Choose capacity at the 83.3rd percentile of the predictive demand distribution under this simple model.

A percentile is a location in that distribution. It is not a percentage increase to the mean. Constraints and other costs can change the decision.

A recommendation for the main ShopEase example

Use seasonal naive as the reference because it had the smallest MAE in the two earlier validation windows.

For the final week issued after day 28, its MAE was ten orders per day. Every daily forecast was too low.

Obtain longer history, compare a method that adapts to changing level, and calibrate uncertainty before applying staffing costs.

Report the target, issue time, horizon and data cutoff with the forecast.

Worked exercises with separate data

Worksheet A uses two training weeks and one later scoring week. The count scale differs from the main example.

Worksheet B checks evaluation design, missing values, uncertainty and capacity decisions.

Compare the methods at the same issue time and over the same horizon.

Worksheet A: the two observed training weeks

Weekday	Week 1 observed	Week 2 observed
Monday	40	50
Tuesday	50	55
Wednesday	50	60
Thursday	60	65
Friday	80	90
Saturday	100	110
Sunday	60	70

Issue all seven week-3 forecasts together at the end of week 2, after day 14.

A1 The worksheet forecast task

State what is forecast, the time interval, when the forecast is issued, how far ahead it goes and the seasonal period.

Describe the training pattern using only weeks 1 and 2.

A1 Seven daily order forecasts after day 14

Target: eligible paid orders each calendar day. Issue time: end of week 2, after day 14.

Targets: days 15 to 21, one to seven days ahead. Seasonal period: seven daily observations.

Training has Saturday peaks and a higher level in week 2. Future actuals have no role in this description.

A2 The two forecast paths

Using only the first 14 days, produce the seven naive and seasonal naive forecasts for week 3.

Identify the exact source observation for Wednesday and Saturday under each method. Calculate each weekly forecast total.

A2 Naive totals 490 and seasonal naive totals 500

Forecast	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Naive	70	70	70	70	70	70	70
Seasonal naive	50	55	60	65	90	110	70

Naive copies week 2 Sunday, 70, for both Wednesday and Saturday.

Seasonal naive copies week 2 Wednesday, 60, and week 2 Saturday, 110.

Naive total = 7 x 70 = 490. Seasonal naive total = 500.

A3 The actual week and its daily errors

Quantity	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Actual week 3	55	60	65	70	95	115	75
Naive forecast	70	70	70	70	70	70	70
Seasonal naive	50	55	60	65	90	110	70

Using actual minus forecast, calculate both seven-day error rows.

A3 Seasonal naive is five too low each day

Error row	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Naive	-15	-10	-5	0	25	45	5
Seasonal naive	5	5	5	5	5	5	5

Naive Monday error: 55 - 70 = -15.
Seasonal naive Monday error: 55 - 50 = +5.

Positive error means the forecast was too low.

A3 Comparing the accuracy summaries

For naive, absolute errors sum to 105, squared errors sum to 3,025 and signed errors sum to 45.

For seasonal naive, every error is five. Calculate MAE, RMSE and mean error for each method.

A3 Seasonal naive has the smaller MAE

Measure	Naive	Seasonal naive
MAE	105 / 7 = 15	35 / 7 = 5
RMSE	Square root of (3,025 / 7) = 20.788	Square root of (175 / 7) = 5
Mean error	45 / 7 = 6.429	35 / 7 = 5

All metrics use orders as the target unit. Carry unrounded intermediate values into the calculations.

A4 Direction, percentage error and evidence

Actual week 3 totals 535. Seasonal naive forecasts 500 and has 35 total absolute error.

Does it systematically underpredict or overpredict? Calculate WAPE and explain one limit of this comparison.

A4 Underprediction with WAPE about 6.542%

$$\text{WAPE} = 100\frac{35}{535} = 6.542\%$$

All errors are +5, so every forecast is too low. The forecast total is short by 35 orders.

One scoring week cannot establish long-run superiority. More issue times and different conditions are needed.

A5 Monday's actual enters Tuesday's forecast

A teammate uses Monday's actual of 55 to forecast Tuesday. Your original Tuesday forecast was issued on the preceding Sunday.

Are these forecasts evaluating the same task?

A5 The updated forecast uses a later information set

The Sunday-issued Tuesday forecast is two steps ahead. The updated forecast after Monday is one step ahead.

Both can be useful, but they have different issue times and available observations.

Compare competing methods under the same operational issue schedule.

B1 An evaluation design with 52 weeks

You have 52 complete weeks of daily history. Every Sunday, all seven forecasts for the coming week must be issued together.

Propose earlier validation windows and a final test block. Explain how training updates and how the test is protected.

B1 One acceptable chronological design

Stage	Example procedure
Earlier validation	Train weeks 1 to 26, forecast week 27, then expand and continue through week 48
Final test block	Reserve weeks 49 to 52 before choosing the method
Weekly issue schedule	Freeze the procedure, then refit on newly observed weeks if that schedule was declared
Scoring	Preserve each seven-day forecast path and score matching actuals

Freeze the rule before testing. Match the weekly issue schedule.

B2 A reliable zero and a failed extract

Tuesday has a reliable recorded count of zero. Wednesday has no record because the extract failed.

What should the dataset initially show? Why would replacing both with zero distort the series?

B2 A reliable zero and a missing observation

Tuesday stays zero because it is a known observation. Wednesday stays missing while the failed extract is investigated.

A zero replacement would invent a demand collapse on Wednesday and could distort fitting and accuracy.

B3 A percentage error with a zero actual

Actual counts are 0 and 5. Forecasts are 2 and 4.

Calculate MAE and state whether ordinary two-day MAPE exists.

B3 The absolute miss is defined on both days

$$e = (-2,\ 1), \qquad \mathrm{MAE} = \frac{2+1}{2} = 1.5$$

Ordinary MAPE is undefined because its first term divides by zero.

Dropping that day silently would describe a different evaluation.

B4 Four-day prediction interval

In the separate random-walk illustration, the latest value is 100. Independent normal future increments have mean zero and standard deviation ten.

Calculate the four-day approximate 95% interval. Why should this formula not be copied automatically to a seasonal model?

B4 The interval is 60.8 to 139.2

$$\sigma_4 = 10\sqrt{4} = 20$$

$$100 \pm 1.96(20) = [60.8,\ 139.2]$$

The formula depends on the stated random-walk error structure. A seasonal method needs uncertainty estimated under its own assumptions.

B5 A capacity rule under unequal costs

Unmet demand costs ten units per order. Unused capacity costs two units per order.

Which predictive percentile is the target under the simple cost model? Explain why this is not an 83.3% uplift.

B5 A predictive percentile rather than an uplift

$$q = \frac{10}{10+2} = 0.8333\ldots$$

Capacity targets the 83.3rd percentile of the predictive demand distribution.

Its numerical order count depends on that distribution. Multiplying a point forecast by 1.833 does not implement this rule.

B6 Selecting and correcting on the final test week

An analyst chooses the best method using final test outcomes, adjusts its forecasts using that week's mean error, and reports the adjusted accuracy.

What is wrong with treating that score as independent evidence of future performance?

B6 The test outcomes influenced development

Both method selection and error correction used the test data. The adjusted result is a tuned score on an inspected week.

Use earlier validation for development, or designate this week as development data and assess the revised procedure on later untouched observations.

Preserve the original forecast record and state the changed evaluation status.

A forecast recommendation for the packing team

Write a short recommendation naming the reference forecast, the missing evidence or uncertainty, and what should be monitored.

Connect the recommendation to daily capacity decisions.

A reference forecast with an operational follow-up

Use seasonal naive as a reference and compare it with a method that adapts to changing level through earlier weekly forecast origins.

Obtain longer history and calibrated predictive uncertainty before translating forecasts into capacity under shortage and surplus costs.

Monitor weekday and horizon-specific errors, persistent underprediction, revisions and missing dates.

Three review questions

1 What information is allowed at a forecast origin?

2 Why can low mean error coexist with large MAE?

3 How does a prediction interval differ from a confidence interval for a mean?

The definitions behind an honest forecast

1 Information actually available when that forecast was issued, under the declared data cutoff.

2 Positive and negative errors can cancel in a signed average. Absolute errors retain the size of each miss.

3 A prediction interval concerns a future observation. A confidence interval for a mean concerns uncertainty in that mean.

The method follows the business question

Business question	Method covered
Did a change improve the metric?	Experiment with a defined outcome
Which products occur together?	Association-rule counts and comparisons
How long until the first event?	Cohorts, censoring and survival analysis
What will the next calendar values be?	Forecasts evaluated at realistic issue times

Every analysis needs a population, comparison and decision.

Optional: a recent-window mean forecast

$$M_T = \frac{1}{k}\sum_{j=0}^{k-1} y_{T-j}$$

For the latest seven ShopEase days, the average is 1,210 / 7 = 172.857. One declared rule repeats that fixed value at every horizon.

A recursive moving window that inserts its own forecasts is a different rule. A centered historical average uses later values and is unavailable in real time.

Choose window length using earlier validation.

Optional: why the ACF differs from a lag-plot correlation

$$r_k = \frac{\sum_{t=k+1}^{T}(y_t - \overline{y})(y_{t-k} - \overline{y})}{\sum_{t=1}^{T}(y_t - \overline{y})^2}$$

This definition uses the full-series mean and denominator, with fewer numerator pairs at larger lags.

The lag-7 pairs lie exactly on y(t) = y(t - 7) + 10, so their separate Pearson correlation is one. The displayed sample ACF at lag 7 is 0.688.

Optional: the weak-stationarity conditions

$$E(Y_t) = \mu, \qquad \mathrm{Var}(Y_t) = \sigma^2 < \infty$$

$$\mathrm{Cov}(Y_t, Y_{t-k}) = \gamma_k$$

The mean and variance stay constant, and covariance depends on lag. The conditions concern a process, not identical observed values.

A flat naive forecast does not prove a stationary underlying series. A random walk provides a counterexample.

Optional: forecast summaries and decision loss

Expected squared error is minimized by the conditional mean. Expected absolute error is minimized by a conditional median.

Asymmetric linear shortage and surplus costs select a predictive quantile under the stated decision model.

Forecast accuracy and operational value can rank methods differently when the business penalizes different errors unequally.

MASE is another comparison scale, using a defined training-data naive error denominator. Its denominator must be nonzero.

Optional spreadsheet setup

A2:A36 contains day indices 1 to 35. B2:B36 contains the corresponding main-example actual counts.

Day 28 is B29. The forecast input range ends there. Days 29 to 35 are later outcomes used for scoring.

D2:D8 contains target days 29 to 35. Columns E to I hold actual, mean, naive, seasonal naive and drift values.

Columns J, K and L hold seasonal error, absolute error and squared error.

Optional spreadsheet forecast formulas

Cell	Formula	Meaning
F2	=AVERAGE($B$2:$B$29)	Mean of 28 training days
G2	=$B$29	Latest observed count
H2	=INDEX($B$2:$B$29,D2-7)	Matching weekday last week
I2	=$B$29+(D2-28)*($B$29-$B$2)/27	Endpoint drift

Fill down through row 8. All forecast inputs stop at day 28.

Optional spreadsheet actuals and scoring

Cell	Formula	Meaning
E2	=INDEX($B$2:$B$36,D2)	Later actual, for scoring only
J2	=E2-H2	Actual minus seasonal forecast
K2	=ABS(J2)	Absolute seasonal error
L2	=J2^2	Squared seasonal error

Fill down through row 8. MAE = AVERAGE(K2:K8). RMSE = SQRT(AVERAGE(L2:L8)).

Mean error = AVERAGE(J2:J8). Locale settings may change formula separators.

Optional spreadsheet result checks

Quantity	Expected result
Seasonal forecasts	130, 150, 160, 170, 190, 230, 180
Actual minus forecast	10, 5, 15, 10, 10, 10, 10
MAE and RMSE	10 and approximately 10.351
Mean error	10
WAPE	100 x SUM(K2:K8) / SUM(E2:E8) = 5.469%

Extending a forecast training range through B36 would leak the final test week into the forecasts.

Reference formulas: the four baselines

$$\text{Mean: } \widehat{y}_{T+h|T} = \frac{1}{T}\sum_{t=1}^{T} y_t$$

$$\text{Naive: } \widehat{y}_{T+h|T} = y_T$$

$$\text{Seasonal naive: } \widehat{y}_{T+h|T} = y_{T+h-m}, \quad 1 \leq h \leq m$$

$$\text{Drift: } \widehat{y}_{T+h|T} = y_T + h\frac{y_T - y_1}{T-1}$$

Reference formulas: forecast errors and accuracy

$$e_i = y_i - \widehat{y}_i$$

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |e_i|, \qquad \mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} e_i^2}$$

$$\mathrm{ME} = \frac{1}{n}\sum_{i=1}^{n} e_i, \qquad \mathrm{WAPE} = 100\frac{\sum |e_i|}{\sum y_i}$$

Use matched periods and horizons. WAPE here assumes nonnegative actual counts with a positive total.

Source notes and reading map

Day 4 Detailed Teaching Notes, 25 pages. Footers map each slide to the relevant original page. All case data and operational costs are synthetic.

The core follows the 150 minute plan. Optional reference slides contain moving averages, formal definitions and spreadsheet formulas.

The worksheet uses separate data. The random-walk prediction intervals are a separate illustration, rather than intervals estimated from the ShopEase counts.

Method references

Hyndman and Athanasopoulos, Forecasting: Principles and Practice, third edition. The supplied notes use these chapters for standard methods.

Chapter 2: Time series graphics
https://otexts.com/fpp3/graphics.html

Chapter 5: The forecaster's toolbox
https://otexts.com/fpp3/toolbox.html

Chapter 9: Stationarity and differencing
https://otexts.com/fpp3/arima.html

The teaching notes supply the numerical datasets, worked cases and exercises.